

Topic 1: Linear Regression and Regularization

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The September 15 lecture connects least squares, prediction error, maximum likelihood, regularization, coordinate descent, and convexity. These notes expand the derivations and provide supporting examples.

1.1 Least squares and the geometry of prediction

1.1.1 Notation and the normal equations

Let $A \in \mathbb{R}^{n \times d}$ be a fixed design matrix, $y \in \mathbb{R}^n$ the response vector, and $\theta \in \mathbb{R}^d$ the coefficient vector. We use θ for the coefficients throughout; this is the optimization variable called x on several handwritten pages. The ordinary least-squares (OLS) problem is

$$\min_{\theta \in \mathbb{R}^d} f(\theta), \quad f(\theta) = \frac{1}{2} \|A\theta - y\|_2^2. \quad (1.1)$$

Expanding the square gives

$$f(\theta) = \frac{1}{2} \theta^\top A^\top A \theta - \theta^\top A^\top y + \frac{1}{2} y^\top y, \quad \nabla f(\theta) = A^\top (A\theta - y).$$

Thus every minimizer satisfies the *normal equations*

$$A^\top A \hat{\theta} = A^\top y. \quad (1.2)$$

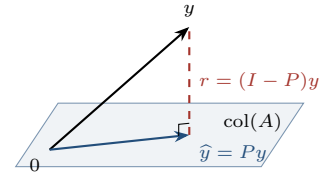
For now assume that A has full column rank d , so $d \leq n$. Then $z^\top A^\top A z = \|Az\|_2^2 > 0$ for every $z \neq 0$, and $A^\top A$ is invertible. The unique solution and its fitted responses are

$$\boxed{\hat{\theta} = (A^\top A)^{-1} A^\top y}, \quad \hat{y} = A\hat{\theta} = Py, \quad P := A(A^\top A)^{-1} A^\top. \quad (1.3)$$

Here $P \in \mathbb{R}^{n \times n}$ is the *hat matrix*. The response y is observed data; it need not equal the noise-free regression mean.

1.1.2 Why the fitted response is a projection

The residual $r = y - \hat{y}$ satisfies $A^\top r = 0$. Therefore, it is orthogonal to every vector in $\text{col}(A)$, while $\hat{y} \in \text{col}(A)$. The diagram shows the decomposition into a fitted component and an orthogonal residual. For any other θ , the cross term in the squared residual vanishes, giving the Pythagorean identity below.



$$\|y - A\theta\|_2^2 = \|y - \hat{y}\|_2^2 + \|\hat{y} - A\theta\|_2^2. \quad (1.4)$$

Thus $\hat{y}$ is the closest point in $\text{col}(A)$ to y .

1.2 Projection matrices: idempotence, eigenvalues, and trace

The OLS hat matrix has two essential properties:

$$P^\top = P, \quad (1.5)$$

$$P^2 = A(A^\top A)^{-1} \underbrace{A^\top A}_{d \times d} (A^\top A)^{-1} A^\top = P. \quad (1.6)$$

The second property is called *idempotence*: projecting a vector already in the target subspace does nothing further. Symmetry together with idempotence makes P an *orthogonal* projection. In particular, $PA = A$.

1.2.1 Eigenvalues are zero or one

Suppose $Pv = \mu v$ for an eigenvector $v \neq 0$. Since $P^2 = P$,

$$P^2 v = \mu^2 v = Pv = \mu v \implies \mu(\mu - 1)v = 0.$$

Consequently, $\mu \in \{0, 1\}$. The eigenvalue 1 corresponds to directions retained by the projection, and 0 to directions removed by it. For the OLS matrix these spaces are $\text{col}(A)$ and its orthogonal complement, respectively.

1.2.2 Computing the trace

For $B \in \mathbb{R}^{n \times d}$ and $C \in \mathbb{R}^{d \times n}$,

$$\text{tr}(BC) = \sum_{i=1}^n \sum_{j=1}^d B_{ij} C_{ji} = \text{tr}(CB).$$

Applying this identity to P gives

$$\text{tr}(P) = \text{tr}((A^\top A)^{-1} A^\top A) = \text{tr}(I_d) = d. \quad (1.7)$$

Equivalently, P has d eigenvalues equal to one and $n - d$ equal to zero. The complementary projection $Q = I_n - P$ satisfies

$$Q^\top = Q, \quad Q^2 = Q, \quad PQ = QP = 0, \quad \text{tr}(Q) = n - d.$$

1.2.3 An idempotent matrix need not be an orthogonal projection

The handwritten example

$$P_{\text{obl}} = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \quad P_{\text{obl}} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u - v \\ 0 \end{pmatrix}$$

satisfies $P_{\text{obl}}^2 = P_{\text{obl}}$, but $P_{\text{obl}}^\top \neq P_{\text{obl}}$. It projects onto the horizontal axis *along the direction* $(1, 1)^\top$, rather than perpendicularly. For example, $(2, 1)^\top$ is mapped to $(1, 0)^\top$. It is an oblique projection. The squared-error identities above rely on the orthogonality of the OLS projector.

If A is rank deficient. Let $r = \text{rank}(A)$. Replace the inverse-based formula by the Moore–Penrose pseudoinverse: $\hat{\theta} = A^\dagger y$ is a minimum-norm least-squares solution and $P = AA^\dagger$ is the unique orthogonal projector onto $\text{col}(A)$. Coefficients may be nonunique, but fitted responses are unique, and $\text{tr}(P) = r$. Formulas below use full column rank; their projection versions remain valid with d replaced by r .

1.3 Noise, training error, and prediction error

1.3.1 The fixed-design linear model

Assume that the data satisfy

$$y = A\theta^* + \varepsilon, \quad \mathbb{E}[\varepsilon] = 0, \quad \text{Cov}(\varepsilon) = \sigma^2 I_n. \quad (1.8)$$

Here θ^* is the true coefficient vector and σ^2 is the noise variance. Independent $\mathcal{N}(0, \sigma^2)$ noise coordinates satisfy these assumptions, but Gaussianity is not needed for the expected squared errors. All expectations in this section are conditional on the fixed matrix A .

Because $\mathbb{E}[\varepsilon_i^2] = \text{Var}(\varepsilon_i) + \mathbb{E}[\varepsilon_i]^2 = \sigma^2$ and $\mathbb{E}[\varepsilon_i \varepsilon_j] = 0$ for $i \neq j$,

$$\mathbb{E}[\varepsilon \varepsilon^\top] = \sigma^2 I_n. \quad (1.9)$$

For any deterministic matrix B with n columns,

$$\begin{aligned} \mathbb{E} \|B\varepsilon\|_2^2 &= \mathbb{E} \text{tr}(B\varepsilon\varepsilon^\top B^\top) \\ &= \text{tr}(B\mathbb{E}[\varepsilon\varepsilon^\top]B^\top) = \sigma^2 \text{tr}(BB^\top). \end{aligned} \quad (1.10)$$

This trace identity is the main tool in both error calculations.

1.3.2 Expected training error

Using $PA = A$ and $\hat{y} = Py$,

$$\hat{y} = A\theta^* + P\varepsilon, \quad y - \hat{y} = (I_n - P)\varepsilon.$$

Define the mean squared training error by $\text{MSE}_{\text{train}} = n^{-1} \|y - \hat{y}\|_2^2$. Then

$$\begin{aligned} \mathbb{E}[\text{MSE}_{\text{train}}] &= \frac{\sigma^2}{n} \text{tr}((I_n - P)(I_n - P)^\top) \\ &= \frac{\sigma^2}{n} \text{tr}(I_n - P) = \boxed{\sigma^2 \left(1 - \frac{d}{n}\right)}. \end{aligned} \quad (1.11)$$

The equality $(I_n - P)^2 = I_n - P$ follows from $P^2 = P$. The unnormalized expected residual sum of squares is $\sigma^2(n - d)$.

Interpretation. Fitting the model removes d directions of the training noise from the residual. The training loss is therefore optimistic as an estimate of future prediction error. For $n > d$, this calculation also yields an unbiased noise-variance estimator:

$$\hat{\sigma}^2 = \frac{\|y - \hat{y}\|_2^2}{n - d}, \quad \mathbb{E}[\hat{\sigma}^2] = \sigma^2. \quad (1.12)$$

If the design includes an intercept, that fitted intercept counts among the d parameters. The identity assumes the mean really lies in $\text{col}(A)$; model misspecification introduces an additional approximation-error term.

1.3.3 Fresh responses at the same design points

Let

$$\tilde{y} = A\theta^* + \tilde{\varepsilon}, \quad \mathbb{E}[\tilde{\varepsilon}] = 0, \quad \text{Cov}(\tilde{\varepsilon}) = \sigma^2 I_n,$$

where $\tilde{\varepsilon}$ is independent of the training noise ε . Keep the predictor $\hat{y}$ fitted to the original data. The new residual is

$$\tilde{y} - \hat{y} = \tilde{\varepsilon} - P\varepsilon.$$

Expanding the square and taking expectations over both noise vectors gives

$$\begin{aligned} \frac{1}{n} \mathbb{E} \|\tilde{y} - \hat{y}\|_2^2 &= \frac{1}{n} \left(\mathbb{E} \|\tilde{\varepsilon}\|_2^2 - 2\mathbb{E}[\tilde{\varepsilon}^\top P\varepsilon] + \mathbb{E} \|P\varepsilon\|_2^2 \right) \\ &= \frac{1}{n} (n\sigma^2 - 0 + \sigma^2 \text{tr}(PP^\top)) \\ &= \boxed{\sigma^2 \left(1 + \frac{d}{n} \right)}. \end{aligned} \tag{1.13}$$

Independence and zero means make the cross term zero. Notice that the two noise vectors are different: using the training noise in both places would recover the training-error calculation instead.

1.3.4 Three distinct quantities

Prediction of the noise-free mean has a different error from prediction of a fresh noisy response:

$$\frac{1}{n} \mathbb{E} \|\hat{y} - A\theta^*\|_2^2 = \frac{1}{n} \mathbb{E} \|P\varepsilon\|_2^2 = \frac{\sigma^2 d}{n}.$$

Quantity	Expected mean squared error
Training responses: $n^{-1} \ y - \hat{y}\ _2^2$	$\sigma^2(1 - d/n)$
Noise-free mean: $n^{-1} \ A\theta^* - \hat{y}\ _2^2$	$\sigma^2 d/n$
Fresh responses, same design: $n^{-1} \ \tilde{y} - \hat{y}\ _2^2$	$\sigma^2(1 + d/n)$

The expected gap between fresh-response and training error is $2\sigma^2 d/n$. For instance, $n = 100$, $d = 10$, and $\sigma^2 = 1$ give 0.9, 0.1, and 1.1 for the three rows above.

Scope of the test-error formula. Equation (1.13) is a *fixed-design* result, not a formula for arbitrary unseen covariates. At a new fixed feature vector $a_0 \in \mathbb{R}^d$ with an independent response $y_0 = a_0^\top \theta^* + \varepsilon_0$, the corresponding error is

$$\mathbb{E}[(y_0 - a_0^\top \hat{\theta})^2] = \sigma^2 + \sigma^2 a_0^\top (A^\top A)^{-1} a_0.$$

This follows from $\text{Cov}(\hat{\theta}) = \sigma^2 (A^\top A)^{-1}$.

When do residuals sum to zero? The normal equations imply $A^\top r = 0$. If $\mathbf{1}_n \in \text{col}(A)$, as when an intercept is included, then $\sum_i r_i = \mathbf{1}_n^\top r = 0$. Without that condition, a zero residual sum is not guaranteed.

1.4 Maximum likelihood and squared-error loss

1.4.1 A coin-flipping example

Suppose a coin has head probability p . In $n = 100$ independent flips, we observe $k = 70$ heads. The likelihood of this count is

$$L(p) = \binom{100}{70} p^{70} (1-p)^{30}, \quad 0 \leq p \leq 1.$$

The binomial coefficient counts the sequences with 70 heads and is constant with respect to p . Since logarithm is strictly increasing, maximizing the likelihood is equivalent to maximizing

$$\ell(p) = 70 \log p + 30 \log(1-p), \quad 0 < p < 1.$$

Its derivatives are

$$\ell'(p) = \frac{70}{p} - \frac{30}{1-p}, \quad \ell''(p) = -\frac{70}{p^2} - \frac{30}{(1-p)^2} < 0.$$

The unique maximum satisfies $70(1-p) = 30p$, so $\hat{p} = 0.7$. Generally, $\hat{p} = k/n$; the cases $k = 0$ and $k = n$ attain their maxima at the boundary. If the exact ordered sequence is observed instead of its head count, the binomial coefficient is absent and the maximizer is unchanged.

1.4.2 Independent Gaussian noise yields least squares

Write the i th row of A as $a_i^\top$. Under

$$y_i = a_i^\top \theta + \varepsilon_i, \quad \varepsilon_i \text{ independently distributed as } \mathcal{N}(0, \sigma^2),$$

the conditional density is

$$p(y_i | a_i, \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - a_i^\top \theta)^2}{2\sigma^2}\right).$$

Conditional independence gives the joint likelihood

$$L(\theta; \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\|y - A\theta\|_2^2}{2\sigma^2}\right). \quad (1.14)$$

For any fixed $\sigma^2 > 0$, the log-likelihood is

$$\ell(\theta; \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|y - A\theta\|_2^2.$$

Its first term is constant in θ , and its second is a negative multiple of squared error. Therefore,

$$\arg \max_{\theta} \ell(\theta; \sigma^2) = \arg \min_{\theta} \frac{1}{2} \|y - A\theta\|_2^2. \quad (1.15)$$

Gaussian noise thus provides a probabilistic motivation for least squares. OLS can still be computed without Gaussian noise; the likelihood interpretation and the optimization problem are distinct statements.

1.5 Nonlinear curves with a linear parameterization

A linear regression model is linear in its *unknown coefficients*; it need not be a straight line as a function of the original input. For scalar inputs $t_1, \dots, t_n$, a cubic model is

$$f_\theta(t) = \theta_1 t^3 + \theta_2 t^2 + \theta_3 t + \theta_4.$$

Define the feature map $\phi(t) = (t^3, t^2, t, 1)^\top$ and design matrix

$$A = \begin{pmatrix} t_1^3 & t_1^2 & t_1 & 1 \\ t_2^3 & t_2^2 & t_2 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ t_n^3 & t_n^2 & t_n & 1 \end{pmatrix}, \quad \theta = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \end{pmatrix}. \quad (1.16)$$

Fitting the cubic is exactly $\min_\theta \frac{1}{2} \|A\theta - y\|_2^2$. The same construction works with fixed polynomial, trigonometric, or other chosen basis functions. It ceases to be linear regression in all parameters if the feature map itself contains additional unknown nonlinear parameters.

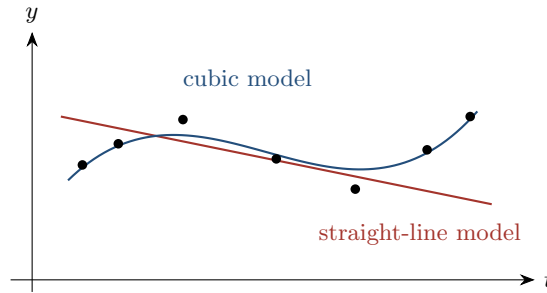


Illustration: both curves are linear in their own coefficients. The drawing is schematic, not the output of a fitted numerical experiment.

1.5.1 Why introduce regularization?

Adding features enlarges the model class and cannot increase the smallest achievable training loss when the feature spaces are nested. However, a flexible model can also fit noise. Nearly dependent feature columns make coefficient estimation unstable because small eigenvalues of $A^\top A$ are inverted.

A regularizer adds a preference among possible fits. Two standard choices are

$$\text{ridge: } \frac{1}{2} \|A\theta - y\|_2^2 + \frac{\lambda}{2} \|\theta\|_2^2, \quad (1.17)$$

$$\text{lasso: } \frac{1}{2} \|A\theta - y\|_2^2 + \lambda \|\theta\|_1. \quad (1.18)$$

Here $\lambda \geq 0$ controls the strength of the penalty. These formulas use a *summed* squared-error loss. If the loss is divided by n , the numerical value of λ must be rescaled to describe the same optimization problem.

1.6 Ridge regression

1.6.1 Closed-form solution and uniqueness

For the ridge objective in (1.17),

$$\nabla f_\lambda(\theta) = A^\top(A\theta - y) + \lambda\theta.$$

Setting the gradient to zero gives

$$(A^\top A + \lambda I_d)\hat{\theta}_\lambda = A^\top y, \quad \boxed{\hat{\theta}_\lambda = (A^\top A + \lambda I_d)^{-1} A^\top y}.$$

For $\lambda > 0$ and any $z \neq 0$,

$$z^\top (A^\top A + \lambda I_d) z = \|Az\|_2^2 + \lambda \|z\|_2^2 > 0.$$

The solution is therefore unique even if A is rank deficient or $d > n$. At $\lambda = 0$ the formula reduces to OLS when A has full column rank.

1.6.2 What does the penalty do?

If $A^\top A = V \text{diag}(s_1^2, \dots, s_d^2) V^\top$, with V orthogonal and $s_j^2 \geq 0$, then

$$\hat{\theta}_\lambda = V \text{diag}\left(\frac{1}{s_1^2 + \lambda}, \dots, \frac{1}{s_d^2 + \lambda}\right) V^\top A^\top y.$$

Adding λ limits the amplification associated with small eigenvalues. When $A^\top A = I_d$, the formula becomes especially simple:

$$\hat{\theta}_\lambda = \frac{1}{1 + \lambda} A^\top y.$$

Ridge shrinks the coefficients continuously. Unlike lasso, it does not have an interval of inputs that are all thresholded to zero. Nevertheless, an individual ridge coefficient *can* be exactly zero because of symmetry or the data. For example, with $A = I_2$ and $y = (2, 0)^\top$, the solution is $(2/(1 + \lambda), 0)^\top$.

Under the correctly specified model (1.8),

$$\mathbb{E}[\hat{\theta}_\lambda] = \theta^* - \lambda(A^\top A + \lambda I_d)^{-1} \theta^*.$$

Thus ridge generally introduces bias. Its purpose is to trade some bias for greater stability; it does not guarantee improved prediction for every dataset or every value of λ .

The fitted-response matrix changes. Ridge uses $H_\lambda = A(A^\top A + \lambda I_d)^{-1} A^\top$. This matrix is symmetric but generally not idempotent. Consequently, the OLS error calculation cannot be reused merely by substituting H_λ for P .

Implementation convention. The displayed ridge penalty applies to every coefficient. An intercept is commonly left unpenalized: center the data, or replace λI_d by a diagonal penalty matrix with a zero in the intercept position. Feature scaling also matters because the penalty acts on coefficients. For numerical work, solve the linear system rather than explicitly forming an inverse; QR or SVD methods are appropriate for unregularized least squares.

1.7 Lasso and the geometry of regularization

Lasso uses the ℓ_1 norm,

$$F(\theta) = \frac{1}{2} \|A\theta - y\|_2^2 + \lambda \sum_{j=1}^d |\theta_j|.$$

The function $|u|$ is convex and continuous, but is not differentiable at $u = 0$. That corner leads to a thresholding effect and allows exact zero coefficients.

1.7.1 Penalized and constrained formulations

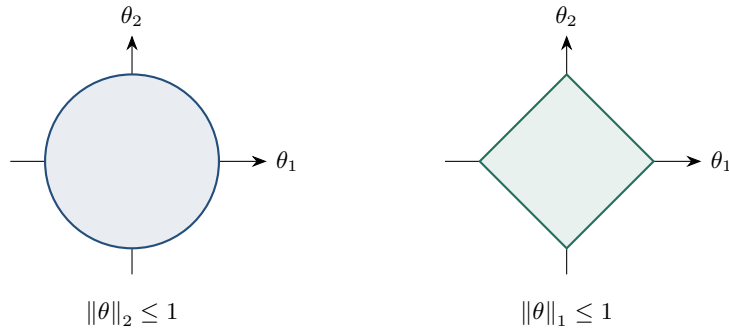
The corresponding constrained problems are

$$\min_{\theta} \frac{1}{2} \|A\theta - y\|_2^2 \quad \text{subject to } \|\theta\|_2^2 \leq t \quad (\text{ridge-type constraint}), \quad (1.19)$$

$$\min_{\theta} \frac{1}{2} \|A\theta - y\|_2^2 \quad \text{subject to } \|\theta\|_1 \leq t \quad (\text{lasso constraint}). \quad (1.20)$$

The parameter t controls the feasible set, whereas λ controls the penalty. They are not numerically equal in general.

If $\hat{\theta}_\lambda$ minimizes a penalized objective $f(\theta) + \lambda g(\theta)$, it also minimizes f subject to $g(\theta) \leq g(\hat{\theta}_\lambda)$. Otherwise, a feasible point with smaller loss would have smaller penalized objective as well. Conversely, under the usual convex constraint qualification, a constrained solution minimizes $f + \mu g$ for a suitable nonnegative Lagrange multiplier μ . For the ridge convention in (1.17), this multiplier is $\mu = \lambda/2$. This correspondence need not be one-to-one, and degenerate boundary cases require care.



In two dimensions, an ℓ_2 ball is a disk, while an ℓ_1 ball is a diamond whose vertices lie on the coordinate axes. When a squared-loss contour first touches an ℓ_1 vertex, one coefficient is zero. This gives geometric intuition for sparsity. It is a tendency, not a guarantee that every lasso solution is sparse or that ridge can never have a zero coefficient.

The next section derives this phenomenon exactly in one dimension. In several dimensions, coordinate descent repeatedly applies that one-dimensional result.

1.8 Soft-thresholding: solving the scalar lasso problem

Consider, for $a \in \mathbb{R}$ and $\lambda \geq 0$,

$$\min_{u \in \mathbb{R}} q(u), \quad q(u) = \frac{1}{2}(u - a)^2 + \lambda|u|. \quad (1.21)$$

The quadratic is strictly convex, so this problem has a unique minimizer. We examine the positive, negative, and zero cases separately.

Case 1: $u > 0$. Here $|u| = u$, so $q'(u) = u - a + \lambda$. The stationary point is $u = a - \lambda$, consistent with $u > 0$ only if $a > \lambda$.

Case 2: $u < 0$. Here $|u| = -u$, so $q'(u) = u - a - \lambda$. The stationary point is $u = a + \lambda$, consistent with $u < 0$ only if $a < -\lambda$.

Case 3: $u = 0$. The left and right derivatives are

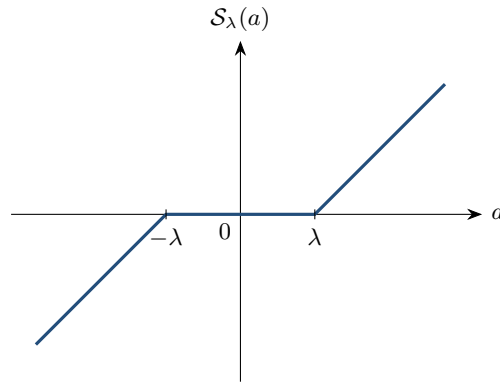
$$q'_-(0) = -a - \lambda, \quad q'_+(0) = -a + \lambda.$$

For a convex function, zero minimizes q if the left derivative is nonpositive and the right derivative is nonnegative. This is equivalent to $|a| \leq \lambda$. Equivalently, $0 \in -a + \lambda\partial|0|$, where $\partial|0| = [-1, 1]$.

Combining the cases gives the *soft-thresholding operator*:

$$u^* = \mathcal{S}_\lambda(a) = \text{sign}(a) \max\{|a| - \lambda, 0\} = \begin{cases} a - \lambda, & a > \lambda, \\ 0, & |a| \leq \lambda, \\ a + \lambda, & a < -\lambda. \end{cases} \quad (1.22)$$

We take $\text{sign}(0) = 0$. Inputs inside the entire interval $[-\lambda, \lambda]$ are mapped to exactly zero; larger inputs are moved toward zero by λ .



Worked values. For $a = 2$, $\lambda = 1$ gives $u^* = 1$ and $q(1) = \frac{1}{2} + 1 = 1.5 < q(0) = 2$. In contrast, $\lambda = 3$ gives $u^* = 0$, because $|a| < \lambda$. The positive-branch stationary point $2 - 3 = -1$ is invalid for that branch. Always check a stationary point against the sign condition used to derive it.

1.9 Coordinate descent for lasso

1.9.1 Reducing one coordinate to a scalar problem

Write $A = [a_1 \ a_2 \ \cdots \ a_d]$, where now $a_j \in \mathbb{R}^n$ denotes a *column* of the design matrix. Thus $A\theta = \sum_j a_j \theta_j$. To update coordinate j , hold the other coordinates fixed and define the partial residual

$$r_j = y - \sum_{k \neq j} a_k \theta_k. \quad (1.23)$$

The terms involving θ_j are exactly

$$\begin{aligned} & \min_{u \in \mathbb{R}} \frac{1}{2} \|a_j u - r_j\|_2^2 + \lambda |u| \\ &= \min_{u \in \mathbb{R}} \frac{1}{2} \|a_j\|_2^2 u^2 - (a_j^\top r_j) u + \lambda |u| + \frac{1}{2} \|r_j\|_2^2. \end{aligned} \quad (1.24)$$

Assume $c_j := \|a_j\|_2^2 > 0$, and put $z_j = a_j^\top r_j$. Completing the square and discarding constants gives

$$\min_u \frac{c_j}{2} \left(u - \frac{z_j}{c_j} \right)^2 + \lambda |u| \iff \min_u \frac{1}{2} \left(u - \frac{z_j}{c_j} \right)^2 + \frac{\lambda}{c_j} |u|.$$

By (1.22), the exact update is

$$\boxed{\theta_j \leftarrow \mathcal{S}_{\lambda/c_j} \left(\frac{z_j}{c_j} \right) = \frac{\mathcal{S}_\lambda(z_j)}{c_j}}. \quad (1.25)$$

The denominator scales both the quadratic center and the threshold. If $\|a_j\|_2^2 = 1$, the update simplifies to $\mathcal{S}_\lambda(a_j^\top r_j)$.

If $a_j = 0$, the data fit does not depend on θ_j . For $\lambda > 0$, set $\theta_j = 0$; for $\lambda = 0$, any value is optimal for that coordinate. This avoids dividing by zero in (1.25).

1.9.2 A residual-based implementation

Maintain the full residual $r = y - A\theta$. Since $r_j = r + a_j \theta_j$, a coordinate update needs no recomputation of the full matrix-vector product:

1. Initialize $\theta = 0$, $r = y$, and $c_j = \|a_j\|_2^2$.
2. For each sweep, visit $j = 1, \dots, d$ using the most recently updated values of all coordinates.
3. Store $u_{\text{old}} = \theta_j$. If $c_j > 0$, compute $z_j = a_j^\top r + c_j u_{\text{old}}$ and $u_{\text{new}} = \mathcal{S}_\lambda(z_j)/c_j$.
4. Set $\theta_j = u_{\text{new}}$ and update $r \leftarrow r - a_j(u_{\text{new}} - u_{\text{old}})$.
5. Repeat until an optimality check or a suitable convergence tolerance is satisfied. Handle zero columns separately as above.

For dense data, one full sweep costs $O(nd)$ arithmetic operations.

1.9.3 What the descent argument proves

Let $\theta^{(k,0)}$ denote the coefficients at the beginning of sweep k , and $\theta^{(k,j)}$ the coefficients after updating coordinate j . An exact coordinate minimization cannot increase the full objective:

$$F(\theta^{(k,0)}) \geq F(\theta^{(k,1)}) \geq \dots \geq F(\theta^{(k,d)}) = F(\theta^{(k+1,0)}). \quad (1.26)$$

Because $F \geq 0$, its values converge to a finite limit. This is the precise conclusion of the descending chain in the handwritten notes. A nonincreasing bounded sequence of objective values, by itself, does not prove convergence of the coefficient vectors or attainment of a global minimum.

1.9.4 Checking global optimality for lasso

The lasso objective is convex. Its necessary and sufficient first-order optimality condition is

$$0 \in A^\top(A\theta - y) + \lambda \partial \|\theta\|_1. \quad (1.27)$$

Using $r = y - A\theta$, this separates into one condition per coordinate:

$$\begin{cases} a_j^\top r = \lambda \operatorname{sign}(\theta_j), & \theta_j \neq 0, \\ |a_j^\top r| \leq \lambda, & \theta_j = 0. \end{cases} \quad (1.28)$$

A point minimizing every coordinate subproblem satisfies these conditions simultaneously and is therefore a global minimizer. Convexity and the separable ℓ_1 penalty are essential to this argument; coordinatewise minimality need not imply global optimality for a general objective.

A practical violation measure is

$$v_j = \begin{cases} |a_j^\top r - \lambda \operatorname{sign}(\theta_j)|, & \theta_j \neq 0, \\ \max\{|a_j^\top r| - \lambda, 0\}, & \theta_j = 0. \end{cases}$$

One may stop when $\max_j v_j$ is below a chosen absolute or appropriately scaled tolerance. A small change in objective is useful to monitor, but can occur before the optimality conditions are well satisfied.

Existence and uniqueness. For $\lambda > 0$, the lasso objective obeys $F(\theta) \geq \lambda \|\theta\|_1$, so its level sets are bounded and a minimizer exists. If A has full column rank, the squared-loss term is strictly convex, hence the minimizer is unique. Without full column rank, uniqueness is not guaranteed, although it can still hold for particular data. Distinguish convergence of objective values, satisfaction of optimality conditions, and uniqueness of coefficients when describing an algorithm's behavior.

Orthogonal-design check. If $A^\top A = I_d$, then

$$\frac{1}{2} \|A\theta - y\|_2^2 = \frac{1}{2} \|\theta - A^\top y\|_2^2 + \text{constant in } \theta.$$

The entire lasso problem separates, and its solution is obtained by applying $\mathcal{S}_\lambda$ to each coordinate of $A^\top y$. For general A , correlated columns couple the coordinates, so this one-shot formula is not valid.

1.10 Convex sets

Definition 1.1 (Convex set) A set $C \subseteq \mathbb{R}^d$ is convex if, for every $x, y \in C$ and every $\alpha \in [0, 1]$,

$$\alpha x + (1 - \alpha)y \in C. \quad (1.29)$$

The points $\alpha x + (1 - \alpha)y$ form the entire line segment between x and y . Convexity asks whether this segment stays in the set, for *every* choice of its two endpoints. The definition does not require the set to be open, closed, bounded, or full-dimensional.

1.10.1 Norm balls are convex

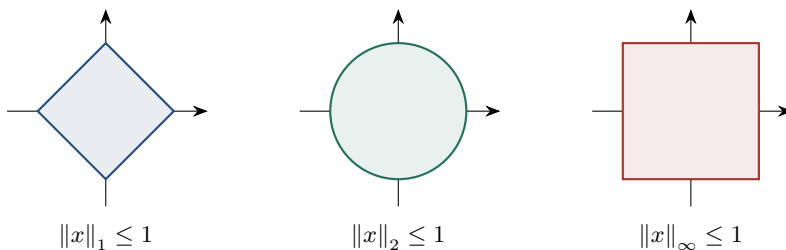
Let $\|\cdot\|$ be any norm and $t \geq 0$. If $\|x\| \leq t$ and $\|y\| \leq t$, the triangle inequality and positive homogeneity imply

$$\begin{aligned} \|\alpha x + (1 - \alpha)y\| &\leq \|\alpha x\| + \|(1 - \alpha)y\| \\ &= \alpha \|x\| + (1 - \alpha) \|y\| \leq t. \end{aligned} \quad (1.30)$$

Thus $\{x : \|x\| \leq t\}$ is convex. In particular, for $1 \leq p < \infty$,

$$\|x\|_p = \left(\sum_{j=1}^d |x_j|^p \right)^{1/p}, \quad \|x\|_\infty = \max_j |x_j|,$$

all corresponding balls are convex. The statement concerns norms; expressions with $0 < p < 1$ are not norms and their unit balls need not be convex.



In $\mathbb{R}^2$, these are a diamond, a disk, and a square. The ℓ_∞ ball is $[-1, 1]^2$ because both coordinates must lie between -1 and 1 .

1.10.2 A ball and its boundary are different sets

The unit sphere $S = \{x : \|x\|_2 = 1\}$ is not convex in $\mathbb{R}^d$ for $d \geq 1$. Both e_1 and $-e_1$ lie in S , but their midpoint is $0 \notin S$. The corresponding *ball* includes the interior and is convex. A sketch of a circle must therefore specify whether it represents the circumference alone or the filled disk. More generally, a visible inward indentation can witness nonconvexity when a segment between two retained points crosses the missing area.

1.10.3 Removing boundary points: two instructive examples

The final board sketches raise a subtle question: does deleting boundary points destroy convexity? The answer depends on which points are removed. The following precise sets illustrate the two situations.

Example 1: a square with its four vertices removed is still convex. Let

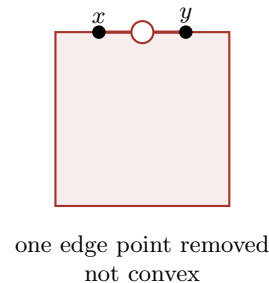
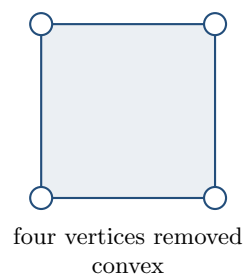
$$C = [0, 1]^2 \setminus \{(0, 0), (0, 1), (1, 0), (1, 1)\}.$$

For $x, y \in C$, their segment lies in $[0, 1]^2$. Could an interior point of the segment equal a deleted vertex? If $\alpha x + (1 - \alpha)y = (0, 0)$ with $0 < \alpha < 1$, nonnegative coordinates force $x = y = (0, 0)$, which is impossible. The same argument, using coordinates bounded above by one as needed, applies to all four vertices. Therefore the segment avoids every removed point, and C is convex. It is neither open nor closed in $\mathbb{R}^2$.

Example 2: removing the midpoint of an edge destroys convexity. Let

$$D = [0, 1]^2 \setminus \{(1/2, 1)\}.$$

The retained points $x = (1/4, 1)$ and $y = (3/4, 1)$ have midpoint $(x + y)/2 = (1/2, 1) \notin D$. This single counterexample proves that D is not convex. Removing an interior point from a disk or square similarly destroys convexity: choose two nearby retained points on opposite sides of the hole.



Open circles denote excluded points; filled regions and the other boundary points are retained. These examples show why convexity must be checked with line segments, rather than inferred only from whether a boundary is included.

1.10.4 Other examples used in optimization

The entire space $\mathbb{R}^d$, affine subspaces, and halfspaces are convex. For example, if $b^\top x \leq c$ and $b^\top y \leq c$, then $b^\top(\alpha x + (1 - \alpha)y) \leq c$. Intersections of convex sets are convex because a segment between two points in the intersection stays in every set. This explains the convexity of feasible regions defined by several linear inequalities and norm bounds.

1.11 Connecting the examples to convex optimization

The following definition completes the connection between the final convexity sketches and the regression objectives studied earlier.

Definition 1.2 (Convex function) Let C be a convex set. A function $f : C \rightarrow \mathbb{R}$ is convex if, for all $x, y \in C$ and $\alpha \in [0, 1]$,

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y). \quad (1.31)$$

Geometrically, the graph lies below the chord between any two graph points. The functions u^2 and $|u|$ both satisfy the definition, even though $|u|$ is not differentiable at zero. Convexity does not require differentiability.

1.11.1 Why the regression problems are convex

For squared-error loss, the Hessian is

$$\nabla^2 f(\theta) = A^\top A \succeq 0, \quad z^\top A^\top A z = \|Az\|_2^2 \geq 0.$$

Thus least squares is convex, and it is strictly convex when A has full column rank. Ridge adds the convex quadratic $\frac{\lambda}{2} \|\theta\|_2^2$; for $\lambda > 0$ its Hessian is positive definite regardless of the rank of A . Lasso adds a nonnegative multiple of a norm, which is convex by the same triangle inequality argument used for norm balls. A sum of convex functions is convex.

The constrained formulations (1.19)–(1.20) also have convex feasible sets. For ridge, the constraint is the ball $\|\theta\|_2 \leq \sqrt{t}$ when $t \geq 0$; for lasso it is an ℓ_1 ball. Therefore both are convex optimization problems.

1.11.2 Why a local minimum is global

Suppose x is a local minimum of a convex function on a convex feasible set, but a feasible y satisfies $f(y) < f(x)$. For sufficiently small $t > 0$, $x_t = (1 - t)x + ty$ is feasible and arbitrarily close to x , while convexity gives

$$f(x_t) \leq (1 - t)f(x) + tf(y) < f(x),$$

contradicting local minimality. Thus every local minimum is global. Strict convexity additionally gives *at most one* minimizer; existence must still be established separately.

1.11.3 Review questions

1. Prove that $\hat{y}^\top (y - \hat{y}) = 0$. Which property of P makes the proof work?
2. Explain why the fresh-response error contains $+d/n$, whereas training error contains $-d/n$. Where is independence used?
3. For $\min_u \frac{1}{2}(u + 4)^2 + 1.5|u|$, find the minimizer by checking the scalar cases. Then verify it with soft-thresholding.
4. Derive the coordinate update when the squared-loss term is divided by n . How does the threshold change if λ is kept fixed?
5. Is a disk with one boundary point removed convex? Is a disk with its center removed convex? Give segment-based arguments.